

The Clover Term in QC₂D

Or potentially a novel rapid method of developing a permanent brain injury.

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This is a brief derivation of the clover action for an $SU(2)$ HMC code at non-zero chemical potential (QC₂D). Much of it is copy-pasted from the thesis so it should not be interpreted as a publishable work.

As the suggestive name suggestively suggests, QC₂D is a model where the gauge fields are instead elements of $SU(2)$. Thus the generators for the gauge fields are the Pauli matrices rather than the Gell-Mann matrices. Being a simpler theory gives it a number of computation advantages. But the main motivator for us is a little party trick known as the *Pauli-Gürsey* symmetry

$$U^* = \sigma_2 U \sigma_2 \quad (1)$$

which ensures that

$$\det D = \det [\sigma_2 D \sigma_2] = \det [D^*] \in \mathbb{R} \quad (2)$$

allowing simulations at $\mu_q \neq 0$.

1 Background

We are starting with an $SU(2)$ gauge theory where the gauge fields form a Lie algebra belonging to the adjoint representation of $SU(N)$.

$$A_\mu(x) = \sum_{i=1}^{N^2-1} T_i A_\mu^{(i)}(x) \quad (3)$$

where g is the coupling constant, T_i form a basis for $\mathfrak{su}(N)$ and $A_\mu^{(i)}(x)$ are real-valued fields.

After discretisation we can no longer have infinitesimal gauge transformations. The solution is to instead express the gauge fields as elements of the Lie *group*

$$U_\mu(x) = e^{i g A_\mu(x)} \quad (4)$$

where x is the lattice site and μ is the direction of the link containing the gauge field. These gauge fields are written down as

$$SU(2) = \left\{ \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix}; \alpha, \beta \in \mathbb{C}; |\alpha|^2 + |\beta|^2 = 1 \right\}. \quad (5)$$

Using symmetries, we only need to store the top line of the matrix to carry out the simulation. This is by no means a unique representation of $SU(2)$ matrices. If we consider $\alpha = a + di$ and $\beta = c + bi$ then it is not difficult to see that

$$U_\mu(x) = a\mathbb{1} + ib\sigma_x + ic\sigma_y + id\sigma_z. \quad (6)$$

where $\sigma_x, \sigma_y, \sigma_z$ are the Pauli matrices. This is equivalent to expressing the gauge field as a quaternion. Written in this form it is clear that the real component of α corresponds to the trace of the gauge link and the remaining components to the traceless components of the gauge links. The analysis code uses this latter form to represent the gauge fields.

By convention, we assume that $\mu \geq 0$. But there is no reason that one cannot have a gauge link pointing in the negative direction. In fact, we can even express them in terms of a gauge link in the forwards direction

$$U_{-\mu}(x) = U_\mu^\dagger(x - \hat{\mu}). \quad (7)$$

Closed paths on the lattice correspond to gauge invariant quantities, and thus observable quantities. The plaquette is the simplest closed loop on the lattice and is formed by a square of gauge links

$$U_{\mu,\nu}(x) = U_\mu(x)U_\nu(x + \hat{\mu})U_{-\mu}(x + \hat{\mu} + \hat{\nu})U_{-\nu}(x + \hat{\nu}) \quad (8)$$

or using equation (7)

$$U_{\mu,\nu}(x) = U_\mu(x)U_\nu(x + \hat{\mu})U_\mu^\dagger(x + \hat{\nu})U_\nu^\dagger(x). \quad (9)$$

Usually, we assume that $\nu > \mu$ when calculating the plaquette. This is purely a convention though.

2 Wilson Action

Now that we have plaquettes, they can be used to construct the Wilson gauge action.

$$S_G[U] = \frac{\beta}{N_c} \sum_{x \in \Lambda} \sum_{\nu > \mu} \text{Re Tr} [\mathbb{1} - U_{\mu\nu}(x)] \quad (10)$$

where Λ means all sites on the lattice and

$$\beta = \frac{2N_c}{g^2} \quad (11)$$

is the inverse gauge coupling. The $\mathbb{1}$ term ensures that the action is zero when the gauge fields are set to the identity. In this case $A_\mu(x) = 0$ so corresponds to free fields.

If we naïvely try to discretise the fermion component of the action, we run into the doubling problem. We get around this by using the Wilson Fermion action. The aim is to add a mass term to the propagator that cancels with the doublers yet also disappears in the continuum.

$$\tilde{D}(p) = m + \frac{i}{a} \sum_{\mu=1}^4 \gamma_\mu \sin(p_\mu a) + \frac{1}{a} \sum_{\mu=1}^4 (1 - \cos(p_\mu a)). \quad (12)$$

In the continuum limit, the final term becomes infinitely heavy so decouples from the theory. Additionally, it is 0 for $p = 0$ but non-zero otherwise, and therefore eliminates all the double poles. Inverting the Fourier transform then leaves us with the Wilson-Dirac operator

$$D(x|y) = \sum_{\mu=1}^4 \frac{(1 - \gamma_\mu) U_\mu(x) \delta_{x+\hat{\mu},y} - (1 + \gamma_\mu) U_{-\mu}(x) \delta_{x-\hat{\mu},y}}{2a} + \left(m + \frac{4}{a}\right) \mathbb{1}_{\delta_{x,y}}. \quad (13)$$

If we try to take the massless limit again, we are left with a factor of $\frac{4}{a}$ in the Wilson-Dirac operator, and thus the discretisation errors are now $\mathcal{O}(a)$.

3 Wilson HMC Update

3.1 Gauge Update

The gauge field updates require evaluating

$$U_\mu(x, \tau + \delta\tau) = e^{i\delta\tau \vec{\pi}_\mu(x, \tau)} U_\mu(x) \quad (14)$$

and since $SU(2) \cong SO(3)$ we can evaluate the exponential analytically in a similar manner as we do with complex numbers. Given an $SU(2)$ element written in quaternion form $A = a + \vec{b}$.

$$e^{iA} = e^{ia} \left(\cos(|\vec{b}|) + i\hat{b} \sin(|\vec{b}|) \right). \quad (15)$$

This follows from the Taylor series of $e^{\vec{b}}$. Even powers of $\vec{b}$ are just products of dot products so scalar, whereas odd powers are given by $|\vec{b}|^{n-1} \vec{b} = |\vec{b}|^n \hat{b}$. Since $\vec{\pi}$ doesn't have a scalar

component we can drop the e^a term out front. Letting $\pi = \pi_\mu (x + \frac{\delta\tau}{2})$, the new gauge field is given as

$$U_\mu(x, \tau + \delta\tau) = \delta\tau \begin{pmatrix} \cos |\vec{\pi}| + i\pi_z \sin |\vec{\pi}| & \pi_y \sin |\vec{\pi}| + i\pi_x \sin |\vec{\pi}| \\ -\pi_y \sin |\vec{\pi}| + i\pi_x \sin |\vec{\pi}| & \cos |\vec{\pi}| - i\pi_z \sin |\vec{\pi}| \end{pmatrix} U_\mu(x, \tau). \quad (16)$$

All that's left to do is an $SU(2)$ multiplication. The implementation of this is presented `integrate.c`.

3.2 Momentum updates

In order to update the conjugate momenta, we need to evaluate the force terms acting on them. The force is given by

$$F[U, \phi] = \sum_{i=1}^{N_c^2-1} T_i \nabla^{(i)} S[U, \phi, \bar{\phi}] = \sum_{i=1}^{N_c^2-1} T_i \nabla^{(i)} \left(S_G[U] + \phi^\dagger (M^\dagger M)^{-1} \phi \right) \quad (17)$$

where $\nabla^{(i)}$ is the derivative with respect to A .

3.2.1 Gauge Force

Using equation (4) we can express the derivative of a function $f(U_\mu(x))$ with respect to the gauge field $U_\mu(x)$ and generator j as

$$\nabla^j f(U_\mu(x)) = \frac{\partial}{\partial \alpha} f[e^{i\alpha T_j} U_\mu(x)]_{\alpha=0}. \quad (18)$$

Then the gauge force is just the derivative of the plaquettes

$$\begin{aligned} T_j \nabla^{(j)} S_G[U] &= \frac{\beta}{N_c} T_j \sum_{x \in \Lambda} \sum_{\mu=1}^4 \text{Tr} \left[-\frac{\partial}{\partial \alpha} e^{i\alpha T_j} U_\mu(x) \sum_{\nu \neq \mu} (A_{\mu,\nu}(x) + A_{\mu,-\nu}(x)) \right] \\ &= \frac{\beta}{N_c} T_j \sum_{x \in \Lambda} \sum_{\mu=1}^4 \text{Tr} \left[-iT_j U_\mu(x) \sum_{\nu \neq \mu} (A_{\mu,\nu}(x) + A_{\mu,-\nu}(x)) \right] \end{aligned} \quad (19)$$

where $A_{\mu\nu}(x)$ is the staple in the $\mu - \nu$ direction. The plaquettes are elements of $SU(N_c)$, so we can use equation (4) to express them in the form of equation (6), or the equivalent for $SU(N_c)$ if $N_c \neq 2$. Since the trace of a sum is the sum of the traces, and $T_j \mathbb{1} = T_j$ is traceless, we are left with

$$T_j \nabla^{(j)} S_G[U] = \frac{2\beta}{N_c} \sum_{x \in \Lambda} \sum_{\mu=1}^4 T_j \text{Tr} \left[T_j \sum_{i=1}^{N_c^2-1} (c_i(x, \mu) T_i) \right] \quad (20)$$

where $c_i(x, \mu)$ is the coefficient of generator i at site x in direction μ , and the factor of 2 from the definition of the generators. Expanding the term inside the trace, all of these products are traceless except for $i = j$. Taking the trace of $T_i T_j = \frac{1}{2} \delta_{ij}$ leaves us with

$$T_j \nabla^{(j)} S_G[U] = \frac{\beta}{N_c} \sum_{x \in \Lambda} \sum_{\mu=1}^4 c_j(x, \mu) T_j. \quad (21)$$

Summing over all j , we see that this is merely the traceless component of the gauge action (up to a factor of i) and can be calculated using the same approach as pure gauge theory.

3.2.2 Fermion Force

The entry of the fermion matrix M at index yx is

$$M_{yx} = \delta_{yx} - \kappa \sum_{\mu=\pm 1}^{\pm 4} \delta_{y,x+\hat{\mu}} (1 + \gamma_{\mu}) U_{x,\mu} \quad (22)$$

and thus

$$(M^{\dagger})_{yx} = (M_{yx})^{\dagger} = \delta_{yx} - \kappa \sum_{\mu=\pm 1}^{\pm 4} U_{y,\mu}^{\dagger} (1 + \gamma_{\mu}) \delta_{y,x+\hat{\mu}}. \quad (23)$$

If we now look at the fermionic contribution to the force we can apply the product rule and obtain

$$\nabla^{(j)} (M^{\dagger} M)^{-1} = (M^{\dagger} M)^{-1} \left[\left(\nabla^{(j)} M^{\dagger} \right) M + M^{\dagger} \left(\nabla^{(j)} M \right) \right] (M^{\dagger} M)^{-1} \quad (24)$$

Using equation (18), the derivative of the fermion matrix at site x in direction μ for colour j is given by

$$\begin{aligned} D_{x\mu j} M_{yx} &= -i\kappa \delta_{y,x+\hat{\mu}} (1 + \gamma_{\mu}) T_j U_{x,\mu} \\ D_{x\mu j} M_{yx}^{\dagger} &= i\kappa U_{y,\mu}^{\dagger} T_j (1 + \gamma_{\mu}) \delta_{y,x+\hat{\mu}}. \end{aligned} \quad (25)$$

Applying these derivative terms to equation (24) gives

$$\begin{aligned} (M_{zy})^{\dagger} (D_{x\mu j} M_{zx}) &= \\ & \left[\delta_{yz} - \kappa \sum_{\nu=\pm 1}^{\pm 4} U_{y,\nu}^{\dagger} (1 + \gamma_{\nu}) \delta_{y,x+\hat{\nu}} \right] [-i\kappa \delta_{z,x+\hat{\mu}} (1 + \gamma_{\mu}) T_j U_{x,\mu}] = \\ & -i\kappa \delta_{yz} \delta_{z,x+\hat{\mu}} (1 + \gamma_{\mu}) T_j U_{x,\mu} + i\kappa^2 \sum_{\nu=\pm 1}^{\pm 4} U_{y,\nu}^{\dagger} (1 + \gamma_{\nu} + \gamma_{\mu} + \gamma_{\nu} \gamma_{\mu}) \delta_{y,z+\hat{\nu}} \delta_{z,x+\hat{\mu}} T_j U_{x\mu}. \end{aligned} \quad (26)$$

and

$$\begin{aligned} (D_{x\mu j} (M_{zy})^{\dagger}) M_{zx} &= \\ & [i\kappa U_{y,\mu}^{\dagger} T_j (1 + \gamma_{\mu}) \delta_{y,z+\hat{\mu}}] \left[\delta_{zx} - \kappa \sum_{\nu=\pm 1}^{\pm 4} \delta_{z,x+\hat{\nu}} (1 + \gamma_{\nu}) U_{x,\nu}^{\dagger} \right] = \\ & i\kappa U_{y,\mu}^{\dagger} T_j (1 + \gamma_{\mu}) \delta_{y,z+\hat{\mu}} \delta_{zx} - i\kappa^2 \sum_{\nu=\pm 1}^{\pm 4} U_{y\mu}^{\dagger} T_j (1 + \gamma_{\mu} + \gamma_{\nu} + \gamma_{\mu} \gamma_{\nu}) \delta_{y,z+\hat{\mu}} \delta_{z,x+\hat{\nu}} U_{x\nu}. \end{aligned} \quad (27)$$

The κ^2 terms in both equation (26) and equation (27) are non-zero for $y = z + \hat{\mu} = x + \hat{\mu} + \hat{\nu}$. As we sum over ν and μ we can swap the μ and ν indices and these terms cancel each other out. The κ terms are non-zero for $y = z = x + \hat{\mu}$ in equation (26) and $y = z = x - \hat{\mu}$ in equation (27) respectively. This finally leaves us with an expression for the derivative at site x in direction μ for colour index j

$$\begin{aligned} D_{x\mu j} (M^{\dagger} M)^{-1} &= \\ & (M^{\dagger} M)^{-1}_{xz} \left(-i\kappa (1 + \gamma_{\mu}) T_j U_{x,\mu} + i\kappa U_{x-\hat{\mu},\mu}^{\dagger} T_j (1 + \gamma_{\mu}) \right) (M^{\dagger} M)^{-1}_{zx}. \end{aligned} \quad (28)$$

4 Clover Fermions

The Clover improved Wilson action can be given as

$$S_I = S_{\text{Wilson}} + S_{\text{Clov}} = S_{\text{Wilson}} - \frac{c_{\text{SW}}}{2} \kappa a^5 \sum_{n \in \Lambda} \sum_{\mu < \nu} \bar{\psi}(n) \frac{1}{2} \sigma_{\mu\nu} \hat{F}_{\mu\nu}(n) \psi(n) \quad (29)$$

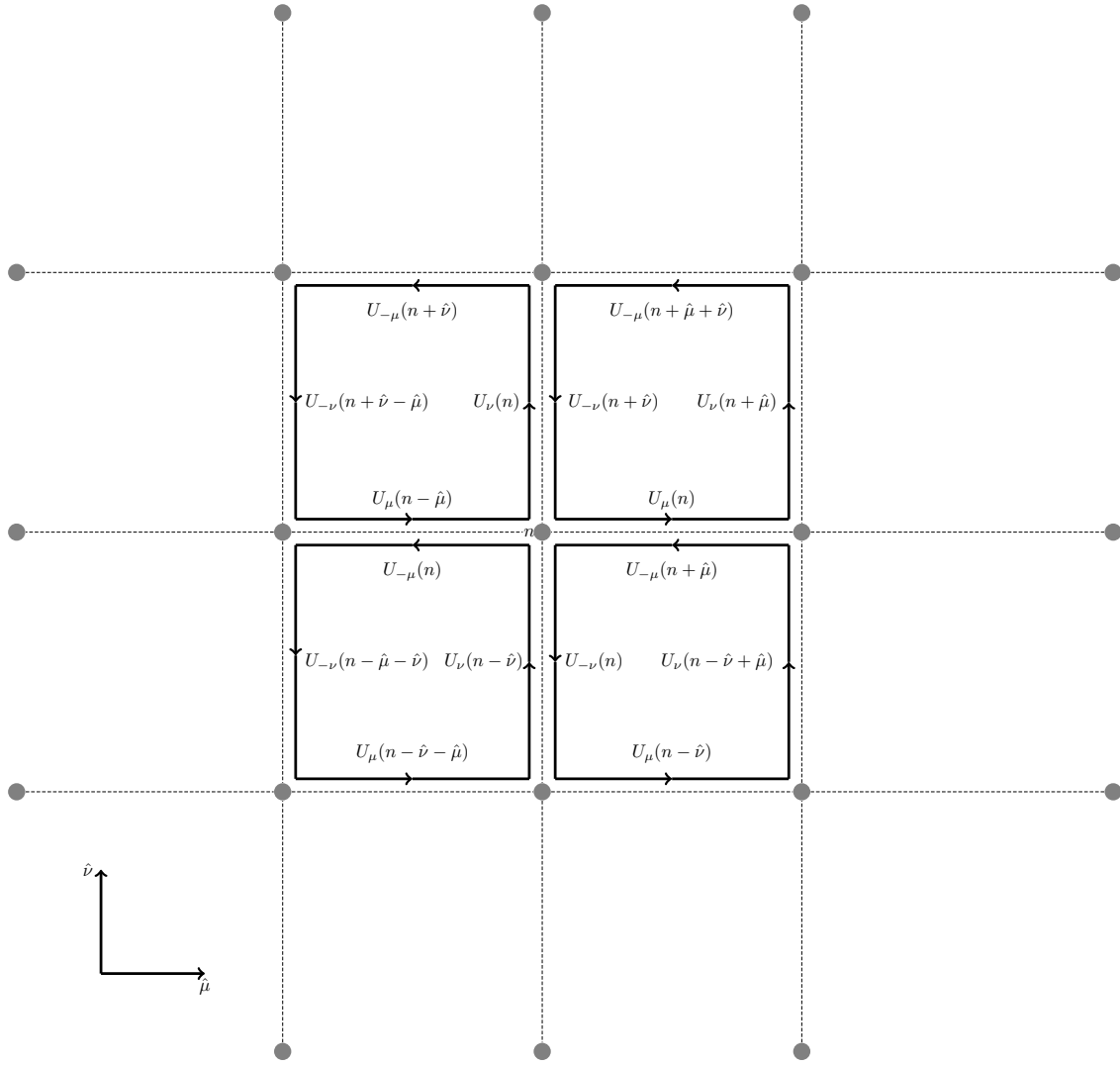


Fig. 1: A half-clover $Q_{\mu\nu}$. The full clover term consists of this minus its hermitian conjugate

where c_{SW} is the clover or Sheiholeslami-Wohlert coefficient, $\sigma_{\mu\nu}$ is the commutator of the Euclidean Dirac matrices in the chiral representation and the clover term at site x in the $\mu - \nu$ direction is given by

$$F_{\mu\nu}(x) = \frac{-i}{8a^2} (Q_{\mu\nu}(x) - Q_{\nu\mu}(x)) \quad (30)$$

where (at least in our implementation, it is not unique)

$$Q_{\mu\nu} = U_{\mu,\nu}(x) + U_{\nu,-\mu}(x) + U_{-\nu,\mu}(x) + U_{-\mu,-\nu}(x) \quad (31)$$

and $U_{\mu,\nu}$ is the plaquette given in equation (8) (see figure 1). Since the $Q_{\mu\nu}$ terms are the sum of four plaquettes they should have the same form as U_{μ} given in equation (5) (but are *not* an element of $SU(2)$), from here it follows that

$$Q_{\mu\nu} - Q_{\nu\mu} = \begin{pmatrix} \alpha - \bar{\alpha} & \beta + \bar{\beta} \\ -\bar{\beta} - \beta & \bar{\alpha} - \alpha \end{pmatrix} = \begin{pmatrix} 2di & 2c + 2bi \\ -2c + 2bi & -2di \end{pmatrix} \quad (32)$$

which is anti-Hermitian. Multiplying out front by $-i$ gives us

$$F_{\mu\nu}(x) = \frac{1}{8a^2} \begin{pmatrix} 2d & 2b - 2ci \\ 2b + 2ci & -2d \end{pmatrix}. \quad (33)$$

which is hermitian and traceless (i.e. of the same form as an element of the Lie Algebra).

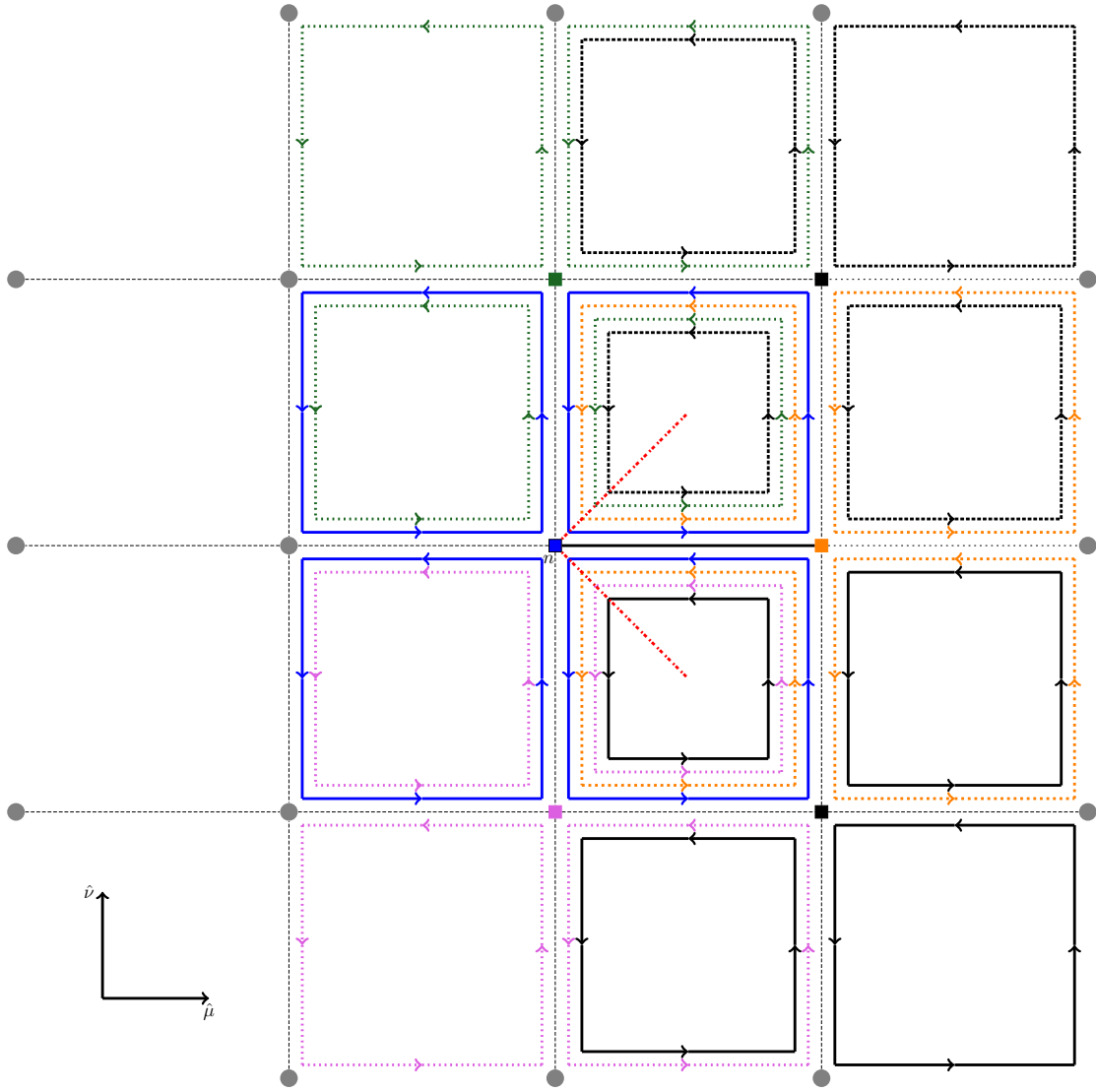


Fig. 2: Leaves containing the link $U_\mu(x)$ or $U_\mu^\dagger(x + \hat{\mu})$ centred on the rectangles. Note that all the clovers drawn in the diagram are $Q_{\mu\nu}$. The contribution from $Q_{\nu\mu}$ is not shown.

5 Clover Force Update

For the clover force update with respect to a link U_μ is similar to the procedure described in section 3.2.2. Only the leaves containing $U_\mu(x)$ or $U_\mu^\dagger(x + \hat{\mu})$ contribute. It does have the additional complication that there are six clovers containing the gauge link between x and $x + \hat{\mu}$ or the conjugate link between $x + \hat{\mu}$ and x .

1. The clover at the site x itself contains the link in the $+\mu, +\nu$ leaf and the $+\mu, -\nu$ leaf.
2. The clover at the site $x + \hat{\mu}$ contains the link in the $-\mu, +\nu$ leaf and the $-\mu, -\nu$ leaf.
3. The clover at the site $x + \hat{\nu}$ contains the link in the $+\mu, -\nu$ leaf.
4. The clover at the site $x - \hat{\nu}$ contains the link in the $+\mu, +\nu$ leaf.
5. The clover at the site $x + \hat{\mu} + \hat{\nu}$ contains the link in the $-\mu, -\nu$ leaf.
6. The clover at the site $x + \hat{\mu} - \hat{\nu}$ contains the link in the $-\mu, +\nu$ leaf.

Unlike the Wilson case where the generator multiplies the plaquette from the left, the location of the generator in the clover force depends on the location of the link in each clover. For the $Q_{\mu\nu}$ leaves seen in figures 2 and 3.

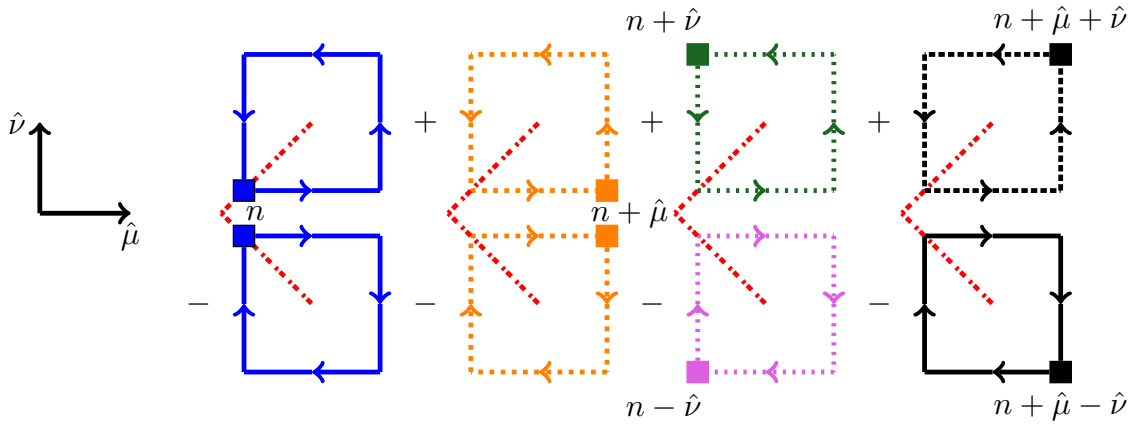


Fig. 3: The components of the clover force with respect to $U_\mu(x)$ (the link with arrow pointing *away* from the centre of the respective clover.) Negative terms are from $Q_{\nu\mu}(x)$ terms. Diagonal lines indicate where the generator appears in the derivative.

1. The clover at the site x itself has the generator multiplying its top-right leaf from the left (w.r.t. $U_\mu(x)$) and bottom right leaf from the right (w.r.t. $U_\mu^\dagger(x + \hat{\mu})$).
2. The clover at the site $x + \hat{\mu}$ has the generator appearing between the third and fourth link of its top left leaf $(\nu, -\mu)$ (w.r.t. $U_\mu(x)$) and between the first and second link of its bottom left leaf $(-\mu, -\nu)$ (w.r.t. $U_\mu^\dagger(x + \hat{\mu})$).
3. The clover at the site $x + \hat{\nu}$ has the generator appearing between the first and second link (w.r.t. $U_\mu(x)$).
4. The clover at the site $x - \hat{\nu}$ has the generator appearing between the third and fourth link (w.r.t. $U_\mu^\dagger(x + \hat{\mu})$).
5. The clover at the site $x + \hat{\mu} + \hat{\nu}$ has the generator appearing between second and third link (w.r.t. $U_\mu(x)$).
6. The clover at the site $x + \hat{\mu} - \hat{\nu}$ has the generator appearing between second and third link (w.r.t. $U_\mu^\dagger(x + \hat{\mu})$).

Looking at the $F_{\mu\nu}(x)$ and the derivatives with respect to $U_\mu(x)$, we see that the derivative from each generator takes the form of an $SU(2)$ group element

$$\begin{aligned} \frac{\partial U_{\mu\nu}(x)}{\partial U_\mu(x)} &= i\sigma_x U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x) \\ &= \begin{pmatrix} -i\bar{\beta} & i\bar{\alpha} \\ i\alpha & i\beta \end{pmatrix} = \begin{pmatrix} -b - ci & d + ai \\ -d + ai & -b + ci \end{pmatrix} \end{aligned} \quad (34)$$

$$\begin{aligned} \frac{\partial U_{\mu\nu}(x)}{\partial U_\mu(x)} &= i\sigma_y U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x) \\ &= \begin{pmatrix} \bar{\beta} & -\bar{\alpha} \\ \alpha & \beta \end{pmatrix} = \begin{pmatrix} c - bi & -a + di \\ a + di & c + bi \end{pmatrix} \end{aligned} \quad (35)$$

$$\begin{aligned} \frac{\partial U_{\mu\nu}(x)}{\partial U_\mu(x)} &= i\sigma_x U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x) \\ &= \begin{pmatrix} i\alpha & i\beta \\ i\bar{\beta} & -i\bar{\alpha} \end{pmatrix} = \begin{pmatrix} -d + ai & -b + ci \\ b + ci & -d - ai \end{pmatrix}. \end{aligned} \quad (36)$$

Let's look at a specific example. Taking $\lambda_j = \sigma_x$ as the generator and using the clover at the site, we see that the contribution from the $Q_{\nu\mu} = Q_{\mu\nu}^\dagger$ leaf

$$\frac{\partial U_{\mu\nu}^\dagger(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} = -iU_\nu(x) U_\mu(x + \hat{\nu}) U_\nu^\dagger(x + \hat{\mu}) U_\mu^\dagger(x) \lambda_j = \left(\frac{\partial U_{\mu\nu}(x)}{\partial U_\mu(x)} \right)^\dagger. \quad (37)$$

Equation (37) is useful as it means we do not have to worry about calculating and keeping track of $U_{\mu\nu}^\dagger(x)$ at the same time as $U_{\mu\nu}(x)$. Just work with the undaggered term instead. We can apply this symmetry to the $U_{-\nu,\mu}(x)$ leaf to obtain $\frac{\partial U_{\mu,-\nu}(x)}{\partial U_\mu(x)}$

$$\frac{\partial U_{-\nu,\mu}(x)}{\partial U_\mu^\dagger(x)} = \left(\frac{\partial U_{-\nu,\mu}^\dagger(x)}{\partial U_\mu(x)} \right)^\dagger \Leftrightarrow \frac{\partial U_{\mu,-\nu}(x)}{\partial U_\mu(x)} = \frac{\partial U_{-\nu,\mu}^\dagger(x)}{\partial U_\mu(x)} = \left(\frac{\partial U_{-\nu,\mu}(x)}{\partial U_\mu^\dagger(x)} \right)^\dagger. \quad (38)$$

Using this symmetry and getting the difference yields

$$\begin{aligned} \frac{\partial F_{\mu\nu}(x)}{\partial U_\mu(x)} &= \frac{-i}{8a} \left(\frac{\partial Q_{\mu\nu}(x)}{\partial U_\mu(x)} - \frac{\partial Q_{\nu\mu}(x)}{\partial U_\mu(x)} \right) \\ &= \frac{-i}{8a} \left(\frac{\partial U_{\mu\nu}(x)}{\partial U_\mu(x)} - \frac{\partial U_{\mu,-\nu}(x)}{\partial U_\mu(x)} \right) = \frac{-i}{8a} \left(\frac{\partial U_{\mu\nu}(x)}{\partial U_\mu(x)} - \left(\frac{\partial U_{-\nu,\mu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} \right)^\dagger \right) \end{aligned} \quad (39)$$

as the total contribution to the derivative of the clover centred at x with respect to $U_\mu(x)$. Repeating for the other contributions

$$\begin{aligned} \frac{\partial F_{\mu\nu}(x + \hat{\mu})}{\partial U_\mu(x)} &= \frac{-i}{8a} \left(\frac{\partial Q_{\mu\nu}(x + \hat{\mu})}{\partial U_\mu(x)} - \frac{\partial Q_{\nu\mu}(x + \hat{\mu})}{\partial U_\mu(x)} \right) \\ &= \frac{-i}{8a} \left(\frac{\partial U_{\nu,-\mu}(x + \hat{\mu})}{\partial U_\mu(x)} - \frac{\partial U_{-\nu,-\mu}(x + \hat{\mu})}{\partial U_\mu(x)} \right) \\ &= \frac{-i}{8a} \left(\frac{\partial U_{\nu,-\mu}(x + \hat{\mu})}{\partial U_\mu(x)} - \left(\frac{\partial U_{-\mu,-\nu}(x + \hat{\mu})}{\partial U_\mu^\dagger(x)} \right)^\dagger \right) \end{aligned} \quad (40)$$

$$\frac{\partial F_{\mu\nu}(x + \hat{\nu})}{\partial U_\mu(x)} = \frac{-i}{8a} \frac{\partial Q_{\mu\nu}(x + \hat{\nu})}{\partial U_\mu(x)} = \frac{-i}{8a} \frac{\partial U_{-\nu,\mu}(x + \hat{\nu})}{\partial U_\mu(x)} \quad (41)$$

$$\begin{aligned} \frac{\partial F_{\mu\nu}(x - \hat{\nu})}{\partial U_\mu(x)} &= \frac{i}{8a} \frac{\partial Q_{\nu\mu}(x - \hat{\nu})}{\partial U_\mu(x)} \\ &= \frac{i}{8a} \frac{\partial U_{\nu,\mu}(x - \hat{\nu})}{\partial U_\mu(x)} = \frac{i}{8a} \left(\frac{\partial U_{\mu,\nu}(x - \hat{\nu})}{\partial U_\mu^\dagger(x)} \right)^\dagger \end{aligned} \quad (42)$$

$$\frac{\partial F_{\mu\nu}(x + \hat{\mu} + \hat{\nu})}{\partial U_\mu(x)} = \frac{-i}{8a} \frac{\partial Q_{\mu\nu}(x + \hat{\mu} + \hat{\nu})}{\partial U_\mu(x)} = \frac{-i}{8a} \frac{\partial U_{-\mu,-\nu}(x + \hat{\mu} + \hat{\nu})}{\partial U_\mu(x)} \quad (43)$$

$$\begin{aligned} \frac{\partial F_{\mu\nu}(x + \hat{\mu} - \hat{\nu})}{\partial U_\mu(x)} &= \frac{i}{8a} \frac{\partial Q_{\nu\mu}(x + \hat{\mu} - \hat{\nu})}{\partial U_\mu(x)} \\ &= \frac{i}{8a} \frac{\partial U_{-\mu,\nu}(x + \hat{\mu} - \hat{\nu})}{\partial U_\mu(x)} = \frac{i}{8a} \left(\frac{\partial U_{\nu,-\mu}(x + \hat{\mu} - \hat{\nu})}{\partial U_\mu^\dagger(x)} \right)^\dagger. \end{aligned} \quad (44)$$

Next, we need to consider the contribution from the derivative with respect to $U_\mu^\dagger(x + \hat{\mu})$. The procedure for this is similar to that for equations (39) to (44). Focusing only on $F_{\mu\nu}(x)$ and using equation (38)

$$\begin{aligned} \frac{\partial F_{\mu\nu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} &= \frac{-i}{8a} \left(\frac{\partial Q_{\mu\nu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} - \frac{\partial Q_{\nu\mu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} \right) \\ &= \frac{-i}{8a} \left(\frac{\partial U_{\nu,-\mu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} - \frac{\partial U_{\nu,\mu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} \right) = \frac{-i}{8a} \left(\frac{\partial U_{\nu,-\mu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} - \left(\frac{\partial U_{\mu,\nu}(x)}{\partial U_\mu(x)} \right)^\dagger \right). \end{aligned} \quad (45)$$

Comparing to equation (39), we can quickly see that

$$\frac{\partial F_{\mu\nu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} = \left(\frac{\partial F_{\mu\nu}(x)}{\partial U_\mu(x)} \right)^\dagger \quad (46)$$

and thus from equation (34) the overall form of the derivatives in the $\hat{\mu}$ direction is

$$\frac{\partial F_{\mu\nu}(x)}{\partial U_\mu(x)} + \frac{\partial F_{\mu\nu}(x)}{\partial U_\mu^\dagger(x + \hat{\mu})} = \frac{-i}{8a} \begin{pmatrix} -i\bar{\beta} + i\beta & i\bar{\alpha} - i\alpha \\ i\alpha - i\bar{\alpha} & i\beta - i\bar{\beta} \end{pmatrix} = \frac{1}{8a} \begin{pmatrix} 2bi & 0 \\ 0 & 2bi \end{pmatrix} \quad (47)$$

which is anti-hermitian. Comparing this with equation (6), we see that the force from the σ_x generator only consists of the σ_x components of the leaves involved in the force. Using a similar approach we can also show that the σ_y and σ_z components behave in the same manner.

Unfortunately, there is no symmetry relating $\frac{\partial F_{\mu\nu}(x)}{\partial U_\mu(x)}$ and $\frac{\partial F_{\nu\mu}(x)}{\partial U_\nu(x)}$ so these must be calculated explicitly. Thankfully this is just a matter of calling the clover force routine with μ and ν swapped.